Aditya Date

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Summary

M.Sc. Mechatronics graduate specializing in sensor fusion and computer vision for autonomous systems. Demonstrable expertise in deep learning for perception, evidenced by a Master's thesis at CARIAD (VW Group) that achieved a record 0.64m RMSE in depth estimation, surpassing the 0.717m KITTI baseline. Proven experience in developing and deploying end-to-end ML pipelines, including YOLOv8 for object detection and multi-modal fusion (LiDAR, Radar). Actively broadening my deep learning expertise beyond Computer Vision to master Generative AI (LLMs, RAG) for multi-modal system design and seeking a full-time role in Sensor Fusion, ADAS, Computer Vision or Robotics.

Skills

- **Programming & Frameworks:** Python, C++, PyTorch, TensorFlow, ROS2, Pandas, OpenCV, LangChain
- **ML & Computer Vision:** Sensor Fusion, Object Detection, Image Processing, Data Engineering, MLOps
- **Tools & Systems:** Linux, Docker, Git, CI/CD Pipelines, Visual Studio Code
- **Soft Skills:** Analytical Thinking, Problem-Solving, Team-Player, Effective Communication, Results-Oriented

Work Experience

- **CARIAD SE (Volkswagen AG)** [View Experience Certificate] — Stuttgart, Germany
  Master Thesis Student. ***Grade: 1.2*** — *Feb 2025 – Oct 2025*

  Research Topic: Road Topography Estimation by Multi-Modal (Lidar, Camera) Perception for Active Suspension.
  - Engineered & benchmarked four deep learning frameworks on a large-scale in-house dataset to generate dense depth maps from **LiDAR** & **camera** inputs.
  - Advanced the SOTA network by integrating **Nvidia** Foundation Stereo, achieving a record **0.64m** RMSE on the in-house dataset and surpassing the KITTI baseline **(0.717m)**.
  - Trained a YOLOv8 model on **6,400** annotated images to detect speedbumps and potholes, achieving **80.5% mAP** on test set of over **9,600** images.
  - Built a sensor data processing & synchronization pipeline, including 3D LiDAR-to-image projection, a **Bosch** anonymization tool (faces/plates), and dense ground truth generation via aggregated LiDAR.

- **Nexustec GmbH** [View Experience Certificate] — Munich, Germany
  Application Engineer Working Student in Computer Vision and Machine Learning. — *Nov 2023 – Jan 2025*
  - Trained GoogleNet models for precise surgical instrument detection (Client: **Hannover Medical School**).
  - Led automated PCB object inspection **(Siemens India)**: deployed QA proof-of-concept for 30+ parts, covering data collection, training, & remote SSH support.
  - Supported deployment of an AI-powered Packaging & Assembly Assistant **(Agile Robots India)** by handling GUI design, data collection, model training, & remote testing.
  - Developed ML models for firefighter helmet classification **(Rosenbauer AG)** to advance safety analytics, collaborated with QA engineer on remote deployment & testing.

Education

- **Hochschule Ravensburg-Weingarten** [View Certificate and Grade Sheet] — Weingarten, Germany
  Master Mechatronics, MSc. — *Sep 2022 – Nov 2025*
  - **GPA:** 1.7
  - **Coursework** - Robotics, Computer Vision, Sensor Fusion, LiDAR and Radar Systems, Automation, Deep - Learning, Embedded Computing, Autonomous Driving, Advanced Mathematics.

- **Mumbai University** [View Degree Certificate] — Mumbai, India
  Bachelor Mechanical Engineering, B.E — *Aug 2018 – May 2022*
  - **GPA:** 1.3
  - **Bachelor Thesis:** *Machine Learning aided Automatic Underwater Structural Inspection Robot.*

Academic Projects

- **LangChain and Llama 3 based Hallucination-Free Offline RAG System** [View GitHub Repo]
  - Built a hallucination-free RAG system deploying a specialized Auditor Agent to enforce strict output compliance.
  - Achieved stable performance on CPU-only infrastructure by optimizing local inference and embedding strategies.
  - Architected a local knowledge engine (FastAPI, ChromaDB) ensuring 100% data sovereignty without external APIs.
- **Multimodal Sensor Fusion for Vehicle Detection and Distance Estimation** [View Project Report]
  - Developed a camera-LiDAR-radar fusion framework with **YOLOv8x** for detecting cars, minivans, & buses.
  - Implemented 3D-to-2D point cloud projection, bounding box filtering, and results visualization.
  - Estimated vehicle distance via geometric algorithms, achieving **1.88m MAE** over 100 test frames.
- **Vehicle Localization and Depth Estimation using mmWave Radar Sensor** [View Project Report]
  - Developed a robust object detection algorithm using a Texas Instruments to track a car up to 50 meters away.
  - Implemented a multi-step filtering method to isolate dynamic points based on relative velocity and estimate car center using dimensional thresholds.
  - Achieved 92.5% detection accuracy against ground truth data with an estimation error below the 2.2m threshold.
- **Object Detection and Depth Estimation using Stereo Camera Sensors** [View Project Report]
  - Designed a Python/YOLOv8 pipeline for vehicle detection & depth estimation using **KITTI** stereo data.
  - Extracted & aligned bounding boxes with ground truth depths, applying IoU filtering & a 2D geometric model.
  - Fused predictions with ground truth, achieving reliable distance estimation **up to 40m**
- **Chest X-Ray Multi-Disease Diagnosis with Deep Learning**
  - Used a pretrained **DenseNet-121 CNN** to concurrently detect 14 different pathologies from chest X-ray images
  - Applied weight normalization to offset low prevalence of certain diseases in the **NIH ChestXray8** dataset (108,000+ X-rays)
  - Incorporated Grad-CAM interpretability to highlight lung regions on the X-rays that influenced each prediction, aiding clinical insight into the model's decisions.

Awards [View Certificate]

- **DAAD Prize 2024:** Awarded for outstanding academic achievements and intercultural engagement.

Publications

- A. Date, Dr. Khatawate et al., *"Numerical Validation of Thrust Produced by Remotely Operated Vehicle,"* in Proceedings of International Conference on Intelligent Manufacturing and Automation, Springer, 2023.[View Publication]
- A. Date, *"Potential Applications and Performance of Energy Recovery in Motor Vehicles: A Literature Review."*, International Research Journal of Engineering and Technology, vol. 8, issue 2, 2021. [View Publication]
- A. Date, Meet Rathod et al., *"Automatic Structural Inspection using Remotely Operated Vehicle,"* International Journal of Design Engineering, vol. 12, no. 2, pp. 87-89, 2023. [View Publication]

Languages

English: Fluent (C1) [View Certificate] **German:** Fluent (C1.1) [View Certificate]

Extracurricular Activities and Social Work [View Testimonials]

- **Member, International Student Council, Hochschule Weingarten** *Feb 2024 – Present*
  - Co-organized & managed logistics for a university-wide Prom Night to integrate international & local students.
  - Led marketing campaigns (social media, university networks), boosting event attendance & engagement.
  - Coordinated event logistics, equipment procurement, & event photography.
- **Student Buddy, Hochschule Weingarten** *Sep 2023 – Present*
  - Mentored new international students during orientation week to facilitate their transition to university.
  - Guided students on website navigation, course registration, & exam procedures.
  - Assisted with administrative tasks (city registration, bank account setup, visa appointments).
- **Student Assistant, Language Center - Hochschule Weingarten** *Nov 2022 – Jan 2023*
  - Developed A1-B2 German learning materials, Moodle exercises & chapter summaries (Netzwerk textbooks).
  - Designed online tests, corrected essays/assessments, and created exam handouts.
  - Promoted DaF examinations campus-wide via marketing campaigns & posters.